

Erdős Problem #805

EP-805

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

For which functions $g(n)$ with $n > g(n) \geq (\log n)^2$ is there a graph on n vertices in which every induced subgraph on $g(n)$ vertices contains a clique of size $\geq \log n$ and an independent set of size $\geq \log n$?

In particular, is there such a graph for $g(n) = (\log n)^3$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>